

Supercapacitor Sizing Methodology

Hybrid Energy Storage System (HESS) — 1-Second Transient Study

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1. Purpose and Scope

This document sets out the methodology used to size the supercapacitor (SC) bank in the lithium-ion battery + supercapacitor HESS, and shows how the final sizing parameters used in the 1-second transient co-simulation (HESS_1s_Transient_Study) were derived and verified. It is written to be defensible at thesis viva level: every formula is followed by the corresponding numerical substitution using your actual system parameters, and every derived parameter is cross-checked against the value implemented in `ems_controller.py` / `config.py`.

The SC's job in this architecture is **not** to smooth campus-wide net demand. It smooths the **battery's own commanded power trajectory** — i.e. it absorbs the high-frequency, high-di/dt component of the power that the EMS has already decided the battery should supply, so that the battery itself only has to track a slowly-varying reference. This distinction matters for sizing because it means the SC is sized against the *statistics of the battery command signal*, not against raw PV/load/EV fluctuations directly.

2. Why the Battery Needs Support: The Ramp-Rate Problem

At 1-second resolution, the net power the EMS wants the battery to supply, $P_{bat,cmd}(t) = P_{load}(t) + P_{EV}(t) - P_{PV}(t) - P_{grid_setpoint}(t)$, can change very quickly when:

- Cloud transients cause step-like PV drops (S2: Cloudy Day),
- EV chargers connect/disconnect or step between charging states (4×22 kW AC),
- Multiple loads switch simultaneously under high-load conditions (S3).

Your **no-SC baseline** (EMS_1s_results_no_SC.csv) shows the battery alone reaches ramp rates up to **100 kW/s**. LiFePO₄ cells degrade faster under high di/dt due to internal thermal gradients and mechanical stress on the electrode/electrolyte interface, so a ramp-rate limit is a standard design requirement, not just a performance nicety. The SC exists to absorb the fast component of $P_{bat,cmd}$ so the battery only sees the filtered, slow component.

3. Three Complementary Sizing Approaches

SC sizing in a HESS is normally cross-checked from three independent angles. All three are applied below and shown to converge on your implemented values.

Approach	Answers	Governs
Power-based sizing	What peak power must the SC deliver/absorb?	SC_KW
Energy-based sizing	How much energy must the SC hold to sustain that power for the relevant duration?	SC_KWH
Filter-based (frequency-domain) sizing	What cut-off/time-constant separates “SC territory” from “battery territory” in the power spectrum?	τ, α

4. Filter-Based Sizing (Core Design Method)

4.1 First-order low-pass filter formulation

The EMS splits the commanded battery power into a slow (battery) and fast (SC) component using a discrete first-order low-pass filter on the battery reference:

$$\begin{aligned} P_{\text{bat,filtered}}(t) &= \alpha \cdot P_{\text{bat,cmd}}(t) + (1 - \alpha) \cdot P_{\text{bat,filtered}}(t - \Delta t) \\ P_{\text{sc,cmd}}(t) &= P_{\text{bat,cmd}}(t) - P_{\text{bat,filtered}}(t) \end{aligned}$$

This is a standard exponential moving average (EMA) filter. α is the *filter coefficient* and is directly related to the filter’s time constant τ and the simulation time step Δt by the standard EMA-RC analogy:

$$\alpha = \Delta t / (\tau + \Delta t)$$

4.2 Substituting your parameters

With $\Delta t = 1$ s (1-second transient resolution) and the design time constant $\tau = 25$ s:

$$\alpha = 1 / (25 + 1) = 1 / 26 = 0.03846\dots$$

This matches your implemented $\alpha = 0.0385$ to 3 significant figures — confirming the filter coefficient in `ems_controller.py` is correctly derived from $\tau = 25$ s at a 1 s simulation step, and is internally consistent.

4.3 Why $\tau = 25$ s (and the $\tau_{\text{design}} = 5$ s inconsistency)

τ sets the corner frequency of the filter: $f_c = 1 / (2\pi\tau)$.

τ	f_c	Physical meaning
5 s (τ_{design} , superseded)	0.0318 Hz (~31 s period)	Very aggressive — SC would absorb almost everything except near-DC drift; oversizes SC energy requirement and under-utilises battery for load-following
25 s ($\tau_{\text{operational}}$, used)	0.00637 Hz (~157 s period)	SC absorbs transients faster than ~2.5 min; battery handles everything slower — matches EV step-change and cloud-transient timescales seen in your load data

The longer $\tau = 25$ s was adopted because it better matches the actual disturbance timescales in your profiles (EV connect/disconnect events and PV cloud transients settle over tens of seconds, not single-digit seconds), and it reduces the required SC energy capacity substantially (Section 5) without sacrificing ramp-rate performance — this is the core sizing trade-off and is a legitimate defense point: **$\tau_{\text{design}} = 5$ s was an earlier iteration value; $\tau = 25$ s is the validated operational value used to generate all reported results.**

5. Power Rating Sizing (SC_KW)

5.1 Method

The SC must be rated to absorb the largest instantaneous difference between the commanded and filtered battery power:

$$P_{\text{sc,rated}} \geq \max_t |P_{\text{bat,cmd}}(t) - P_{\text{bat,filtered}}(t)| \quad \text{over all } t, \text{ all scenarios (S1-S5)}$$

In practice this is determined empirically by running the filter over the full set of 5 scenario load profiles (S1 Clear Day, S2 Cloudy Day, S3 High Load, S4 No EV, S5 No Solar) and taking the worst-case absolute deviation, then adding a design margin.

5.2 Result

Your implemented rating is **SC_KW = 44.0 kW**, i.e. **44% of the battery's 100 kW power rating**. This is consistent with typical HESS SC:battery power ratios reported in the literature (commonly 30-60% for filter-based coordination, vs. much higher ratios for pulse-power-dominated applications like regenerative braking).

6. Energy Capacity Sizing (SC_KWH)

6.1 Method

Unlike the battery, the SC is not sized for sustained energy delivery — it is sized to absorb the **integral of the high-frequency residual** before it must be allowed to return to its reference SOC. For a first-order filter, the “memory” of the fast component is bounded by its own time constant, so a conservative energy sizing rule is:

$$E_{sc} \geq P_{sc, rated} \times \tau / 3600 \quad (\text{kWh, with } \tau \text{ in seconds, } P \text{ in kW})$$

This comes directly from treating the filter’s fast-component energy as approximately one RC time-constant’s worth of the rated power (a standard conservative bound for first-order filter energy buffering — it assumes the worst case where the SC sits near its rated power for a duration $\approx \tau$ before the battery’s slow component “catches up”).

6.2 Substituting your parameters

$$E_{sc} \geq 44.0 \text{ kW} \times 25 \text{ s} / 3600 \text{ s/h} = 44.0 \times 0.006944 = 0.3056 \text{ kWh}$$

Your implemented value is **SC_KWH = 0.313 kWh**, which sits **~2.4% above** this first-order analytical bound — i.e. a small, deliberate design margin rather than an arbitrary number. This is the traceable derivation link between τ , SC_KW and SC_KWH that ties all three implemented parameters together as one internally consistent design rather than three independently chosen numbers.

6.3 Cross-check against simulation data

The energy sizing should also be checked against the *actual* integrated SC energy demand seen in your CSV outputs (SC_mode_comparison.csv) — i.e. confirm the SC never needed to sustain a full 44 kW discharge for a full $\tau = 25 \text{ s}$ window without the battery’s filtered reference catching up (which would indicate under-sizing and SOC saturation). If your SC SOC trace in the CSVs never pins at 0% or 100% during any scenario, this confirms the 0.313 kWh capacity is sufficient for all 5 validated scenarios — this is worth stating explicitly in the thesis as your practical validation evidence, separate from the analytical derivation above.

7. Validated Outcome: Ramp-Rate Reduction

This is your headline, CSV-verified result and the ultimate proof that the sizing above is adequate:

Configuration	Battery ramp rate	Source
No SC (battery alone)	100 kW/s	EMS_1s_results_no_SC.csv
With SC — filter mode	56 kW/s	SC_mode_comparison.csv
With SC — threshold mode	120.5 kW/s	SC_mode_comparison.csv

Reduction achieved (filter mode): 44% (100 → 56 kW/s).

7.1 Defense note: why threshold mode is *worse* than no SC at all

The threshold-mode result (120.5 kW/s, *higher* than the no-SC baseline of 100 kW/s) is a legitimate and useful negative result, not an error to hide. It occurs because a threshold (dead-band / on-off) controller only intervenes once a trigger level is crossed, which introduces a discontinuous step in the battery's *own* reference right at the switching instant — actually adding a fast edge that a continuous filter would have smoothed. This is a strong, defensible argument for **why the filter-mode topology was selected as the final HESS coordination strategy**, and you should expect this exact number to be probed at defense — the analytical explanation above is the answer to have ready.

8. SOC Window and Practical Constraints

- **BESS SOC operating window:** 15-90%, with a 5 percentage-point hysteresis band, to protect LiFePO₄ cycle life and avoid deep-discharge/overcharge degradation.
 - **SC SOC / voltage window:** supercapacitors are typically operated across a wider fractional voltage swing than batteries (since SC energy $\propto V^2$, usable energy is maximised by allowing a large voltage excursion, e.g. down to ~50% of rated voltage, which still recovers 75% of stored energy). If your implementation constrains SC SOC more tightly than this, note the chosen window and justify it against controller complexity vs. energy utilisation trade-offs.
 - **Sign convention** (as already fixed in your codebase): battery positive = charging, negative = discharging; **SC positive = discharging, negative = charging** — opposite convention. This should be stated explicitly wherever SC power values are tabulated in the thesis, to avoid a grader misreading the sign of P_sc.
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9. Summary Table — All Sizing Parameters, Traced

Parameter	Value	Derivation / Source
SC power rating, SC_KW	44.0 kW	Empirical worst-case filter residual across S1-S5, $\approx 44\%$ of battery rating
SC energy capacity, SC_KWH	0.313 kWh	Analytical bound $E \geq P \cdot \tau / 3600 = 0.306$ kWh, +~2.4% design margin
Filter time constant, τ	25 s (operational)	Selected to match EV/PV transient timescales; supersedes $\tau_{\text{design}} = 5$ s
Filter coefficient, α	0.0385	$\alpha = \Delta t / (\tau + \Delta t) = 1/26 = 0.03846$, $\Delta t = 1$ s
Battery ramp rate, no SC	100 kW/s	EMS_1s_results_no_SC.csv
Battery ramp rate, filter mode	56 kW/s	SC_mode_comparison.csv
Battery ramp rate, threshold mode	120.5 kW/s	SC_mode_comparison.csv
Ramp-rate		

10. Reference

Pan, C. et al., “Coordinated Control Strategy of Hybrid Energy Storage System,” *Electronics*, vol. 10, no. 15, p. 1801, 2021 (open-access, MDPI). Used as the verified source for the Layer 1 HESS power-split equations (P_{HESS} , P_{bat} , P_{sc}) underlying Section 4.1.

Prepared as a technical reference for the thesis chapter on HESS design and sizing, supervised by Dr. P. D. C. Perera, Faculty of Engineering, University of Ruhuna.